Reading Sinhala documents
without introducing new mistakes

Fix the mistakes. Keep the correct text.

Not yet qualified | Provisional score: 76.7/100 | No experiments yet

01 The problem in everyday terms

OCR is software that reads text from a scanned page. It can help turn printed Sinhala documents into searchable text. A second AI model may fix reading errors, but it can also rewrite something that was already correct.

For a date, amount or section number, a small change can alter the meaning. The aim is to make correction careful, rather than to rewrite every sentence.

Illustrative number examples - invented, not experimental results

A USEFUL CORRECTION

Image: 2017
OCR: 2O17 → Corrected: 2017

The letter O was mistaken for zero.

A HARMFUL CORRECTION

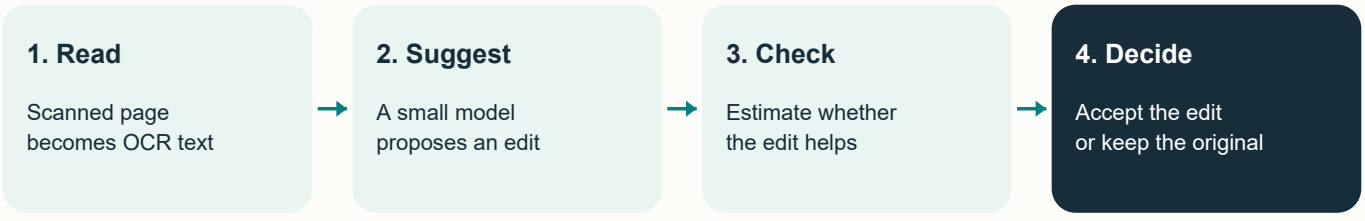
Image: 2017
OCR: 2017 → Changed: 2011

Correct text became wrong.

02 The question we would investigate

Can a small correction model fix remaining Sinhala OCR errors while making fewer unnecessary edits, especially to numbers?

PROPOSED WORKFLOW - NOT A TESTED RESULT



What is the possible gap? Sinhala OCR and automatic correction already exist. The narrower investigation is whether we can choose helpful edits without damaging correct text. That contribution still needs a closer comparison with existing work. [2-4]

A small project we can test

One shared problem. Three independently assessable contributions.

03 Data and practical requirements

Existing dataset: 1,010 scanned Sinhala legislative pages with corrected text. The public release is about 872 MB and states CC BY 4.0. Document IDs allow pages from the same source to stay together when splitting the data. [1]

Resources: start with one fixed OCR engine, cached outputs and small models. Occasional modest GPU access may help. Training time and memory are estimates until measured; no new recordings, surveys or human ratings are planned.

04 How three members contribute research

Member / focus	What to compare	Separate result
A. Better corrections	Learn from actual OCR errors versus dictionary rules and synthetic errors.	Errors fixed versus new errors introduced.
B. When to edit	Train an edit selector versus fixed confidence or edit-distance rules.	Helpful edits at a matched harmful-edit rate.
C. Smaller model	Train a compact corrector with and without extra copying/number preservation.	Accuracy, number matches, memory and speed.

Each member starts from the same unchanged OCR output, so one proposed method does not have to succeed for the others to produce findings. Papers are optional.

05 How we would know whether it works

Measure: character errors, harmful edits and exact matches for numbers.
Compare fairly: unchanged OCR, simple correction rules and stronger models.
Keep tests separate: hold out entire source documents, not random pages.
Check reliability: repeat training and report variation across documents.

First decision: test a small subset before committing. If the starting OCR is already strong, leaving its text unchanged may be the best result. Novelty, actual training costs and baseline-model licences remain unresolved.

SOURCES AND SCOPE

[1] Public dataset: sinhala-ocr-lk-acts-1010
[2] Cross-Temporal Sinhala OCR (2026)
[3] Confident but Wrong (2026): translation editing, not OCR
[4] UCSC/Theekshana Sinhala OCR: existing post-processing

Scope: printed legislative pages; no proven handwriting, phone-photo or operational benefit.